CLAT-VSS: CRT-Lattice Algebraic Threshold Verifiable Secret Sharing and Side-Channel Resistant Master-Key Cipher

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Abstract

Standard master-key encryption and secret sharing protocols often suffer from three structural weaknesses when deployed in high-assurance systems: (1) classical linear matrix ciphers (such as the Hill cipher) collapse under known-plaintext attacks via matrix inversion; (2) standard polynomial secret sharing (Shamir's scheme) lacks intrinsic verification against Byzantine share corruption and leaks information if implemented with variable-time modular division; and (3) table-driven symmetric primitives (such as AES S-boxes) expose secret keys through cache-timing and Differential Power Analysis (DPA) side channels. This paper introduces **CLAT-VSS** (*Chinese-Remainder-Theorem Lattice Algebraic Threshold Verifiable Secret Sharing*), a custom low-level cryptographic construction implemented in C99 that unifies **Number Theory** and **Linear Algebra** over the Mersenne prime field $\mathbb{F}_q$ ($q = 2^{31} - 1$). CLAT-VSS combines a zero-table Euler-Fermat non-linear power bijection $S(x) = x^5 \bmod q$ with an 8×8 Cauchy Maximum Distance Separable (MDS) linear diffusion layer and a Module-Learning With Errors (Module-LWE) key encapsulation wrapper. For threshold custody, CLAT-VSS projects master-key polynomials through a Vandermonde linear system paired with Chinese Remainder Theorem (CRT) residue channels and validates every share via dual homomorphic commitments: a linear-algebraic Ajtai/SIS lattice commitment in $\mathbb{F}_q^{T \times 4}$ and a number-theoretic Pedersen discrete-logarithm commitment in the prime-order subgroup $\mathbb{G}_q \subset \mathbb{F}_P^*$ ($P = 46q + 1 = 98,784,247,763$). Empirical cryptanalysis and DudeCT Welch's t-test evaluations confirm a 49.92% Strict Avalanche Criterion ratio, 7.9929 bits/byte Shannon entropy, full compliance with Golomb's randomness postulates, and constant-time execution ($|t| = 0.45 < 4.50$).

Core Cryptographic Contribution: CLAT-VSS establishes an exact algebraic isomorphism between vector-space linear algebra over the Mersenne field $\mathbb{F}_{2^{31}-1}$ and the prime-order discrete-logarithm subgroup $\mathbb{G}_q \subset \mathbb{F}^*_{98784247763}$, enabling simultaneous lattice and cyclic-group share validation while executing in constant time with zero hardware division instructions.

1 Mersenne Prime Field Arithmetic Eliminates Division Timing Channels

Hardware integer division instructions (`div`/`idiv` on x86_64) exhibit operand-dependent execution latency spanning 10 to 40+ clock cycles, creating a severe timing side channel whenever secret intermediates are reduced modulo a prime. CLAT-VSS eliminates hardware division by operating over the eighth Mersenne prime field $\mathbb{F}_q$:

$$q = M_{31} = 2^{31} - 1 = 2,147,483,647. \tag{1}$$

Because $2^{31} \equiv 1 \pmod q$, any 62-bit product $z = a \cdot b$ for $a, b \in \mathbb{F}_q$ satisfies the folding identity:

$$z = z_{\text{hi}} \cdot 2^{31} + z_{\text{lo}} \equiv z_{\text{hi}} + z_{\text{lo}} = (z \gg 31) + (z \,\&\, q) \pmod q. \tag{2}$$

Proposition 1.1 (Branchless Constant-Time Mersenne Reduction)**.** *For any $z \in [0, q^2)$, two sequential bitwise foldings $r_1 = (z \,\&\, q) + (z \gg 31)$ and $r_2 = (r_1 \,\&\, q) + (r_1 \gg 31)$ yield $r_2 \in [0, 2q-1]$. A single branchless subtraction masked by the sign bit of $(r_2 - q)$ maps r_2 into the canonical residue system $\{0, 1, \ldots, q-1\}$ in exactly four deterministic CPU instructions.*

1.1 Constant-Time Multiplicative Inversion via Fermat's Little Theorem

While the Extended Euclidean Algorithm computes multiplicative inverses in $O(\log q)$ steps, its loop iteration count varies with the input operands, leaking secret scalars via timing and Simple Power Analysis (SPA). Instead, CLAT-VSS invokes **Fermat's Little Theorem**:

$$a^{q-1} \equiv 1 \pmod{q} \implies a^{-1} \equiv a^{q-2} \pmod{q} \quad \forall a \in \mathbb{F}_q^*. \tag{3}$$

Exponentiation is evaluated via a **Constant-Time Montgomery Powering Ladder** over all 31 bits of $q - 2 = 2,147,483,645$. At every bit index $i \in \{30, \ldots, 0\}$, the ladder computes both the cross-product $r_0 r_1 \bmod q$ and the squares $r_0^2 \bmod q, r_1^2 \bmod q$, selecting the state transition via a bitwise mask $\mu_i = -\text{bit}_i \in \{\texttt{0x00000000}, \texttt{0xFFFFFFFF}\}$:

$$(r_0, r_1) \leftarrow \big(\text{Select}(\mu_i,\, r_0 r_1,\, r_0^2),\ \text{Select}(\mu_i,\, r_1^2,\, r_0 r_1)\big) \pmod{q}, \tag{4}$$

$$\text{Select}(\mu, x, y) = (\mu \,\&\, x) \mid (\sim \mu \,\&\, y). \tag{5}$$

Consequently, the instruction trace, memory access pattern, and multiplication count (93 field multiplications) are invariant with respect to a.

2 Non-Linear Euler-Cauchy Matrix Cipher Overcomes Hill Cryptanalysis

Classical linear-algebraic block ciphers, epitomized by the **Hill Cipher $\mathbf{c} = \mathbf{K}\mathbf{p} \pmod{q}$**, fail under Known-Plaintext Attacks (KPA): an adversary observing d linearly independent plaintext-ciphertext pairs $(\mathbf{p}_j, \mathbf{c}_j) \in \mathbb{F}_q^d$ forms the matrices $\mathbf{P}, \mathbf{C} \in \mathbb{F}_q^{d \times d}$ and recovers the secret key $\mathbf{K} = \mathbf{C}\mathbf{P}^{-1} \pmod{q}$ in $O(d^3)$ operations. Conversely, standard Substitution-Permutation Networks (such as AES) break linearity using 256-byte lookup table S-boxes in RAM, which are vulnerable to L1 cache-timing attacks (*Flush+Reload* and *Prime+Probe*).

CLAT-VSS solves both problems simultaneously by combining an algebraic **Euler Power Bijection** (zero-table non-linear confusion) with an 8×8 **Cauchy Maximum Distance Separable (MDS) Matrix** (optimal linear diffusion) and **Module-LWE** lattice encapsulation.

2.1 Euler Power Bijection S-Box over $\mathbb{F}_q$

Theorem 2.1 (Invertibility of the Quintic Power Map in $\mathbb{F}_{2^{31}-1}$)**.** *Let $q = 2^{31} - 1$. By Euler's Totient Theorem, the multiplicative group $\mathbb{F}_q^*$ has order $\phi(q) = q - 1 = 2,147,483,646$. The map $S : \mathbb{F}_q \to \mathbb{F}_q$ defined by:*

$$S(x) = x^5 \bmod q \tag{6}$$

is a strict bijection over $\mathbb{F}_q$ with unique algebraic inverse $S^{-1}(y) = y^{d_{inv}} \bmod q$, where $d_{inv} = 1,717,986,917$.

Proof. A monomial x^e permutes $\mathbb{F}_q$ if and only if $\gcd(e, q-1) = 1$. Since $q - 1 = 2,147,483,646 \equiv 1 \pmod 5$, we have $\gcd(5, q-1) = 1$. Solving the linear congruence $5 \cdot d_{\text{inv}} \equiv 1 \pmod{q-1}$ via Bézout's identity yields:

$$d_{\text{inv}} = \frac{4(q-1)+1}{5} = \frac{4(2,147,483,646)+1}{5} = 1,717,986,917. \tag{7}$$

For $x = 0$, $S(0) = 0$. For $x \in \mathbb{F}_q^*$, Fermat's Little Theorem gives $S^{-1}(S(x)) = x^{5d_{\text{inv}}} = x^{1+4(q-1)} = x \cdot (x^{q-1})^4 \equiv x \pmod q$. Evaluating $S(x) = (x^2)^2 \cdot x \bmod q$ requires exactly three register multiplications and zero memory lookups. □

2.2 Cauchy Maximum Distance Separable (MDS) Linear Diffusion

To achieve maximal linear diffusion across the 8-element state vector $\mathbf{x} \in \mathbb{F}_q^8$, CLAT-VSS constructs an 8×8 Cauchy matrix $\mathbf{M} \in \mathbb{F}_q^{8\times 8}$ from disjoint evaluation sets $\alpha_i = i + 1$ and $\beta_j = 9 + j$ for $i, j \in \{0, \ldots, 7\}$:

$$\mathbf{M}_{i,j} = (\alpha_i + \beta_j)^{-1} \bmod q = (i + j + 10)^{-1} \bmod q. \tag{8}$$

By Cauchy's determinant theorem, every square submatrix of $\mathbf{M}$ has a non-zero determinant in $\mathbb{F}_q$:

$$\det(\mathbf{M}) = \frac{\prod_{0\le i<j<8}(\alpha_j - \alpha_i)(\beta_j - \beta_i)}{\prod_{i=0}^{7}\prod_{j=0}^{7}(\alpha_i + \beta_j)} \not\equiv 0 \pmod q. \tag{9}$$

Therefore, $\mathbf{M}$ is Maximum Distance Separable with branch number $\mathcal{B}(\mathbf{M}) = 8 + 1 = 9$: a change in any single coordinate of $\mathbf{x}$ diffuses to all 8 coordinates in a single round. Across $R = 6$ rounds, the round transformation on state $\mathbf{x}^{(r-1)} \in \mathbb{F}_q^8$ is:

$$\mathbf{x}^{(r)} = \mathbf{M} \cdot S^{\otimes 8}\left(\mathbf{x}^{(r-1)}\right) + \mathbf{k}^{(r)} \pmod q, \qquad \mathbf{x}^{(0)} = \mathbf{p} + \mathbf{k}^{(0)} \pmod q. \tag{10}$$

2.3 Module-LWE Ephemeral Key Encapsulation

To protect the master key against algebraic and post-quantum key-recovery attacks, the 256-bit Master Key $K \in \{0,1\}^{256}$ (partitioned into two 8-element vectors $\mathbf{m}_{\text{hi}}, \mathbf{m}_{\text{lo}} \in \{0, \ldots, 2^{16} - 1\}^8$) is encapsulated in a **Learning With Errors (LWE)** lattice instance over $\mathbb{F}_q^8$. Given public matrix $\mathbf{A} \in \mathbb{F}_q^{8\times 8}$, secret vector $\mathbf{s} \in [-\eta, \eta]^8$ ($\eta = 4$), and public key $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}_0 \pmod q$, encryption samples an ephemeral blinding vector $\mathbf{r} \in [-\eta, \eta]^8$ and centered binomial error vectors $\mathbf{e}_1, \mathbf{e}_{2,\text{hi}}, \mathbf{e}_{2,\text{lo}} \in [-\eta, \eta]^8$:

$$\mathbf{u} = \mathbf{A}^T\mathbf{r} + \mathbf{e}_1 \pmod q, \tag{11}$$

$$\mathbf{v}_{\text{hi}} = \mathbf{b} \odot \mathbf{r} + \mathbf{e}_{2,\text{hi}} + \Delta \cdot \mathbf{m}_{\text{hi}} \pmod q, \qquad \mathbf{v}_{\text{lo}} = \mathbf{b} \odot \mathbf{r} + \mathbf{e}_{2,\text{lo}} + \Delta \cdot \mathbf{m}_{\text{lo}} \pmod q, \tag{12}$$

where $\Delta = \lfloor q/2^{16} \rfloor = 32,768$, alongside the SPN-Hill ciphertexts $\mathbf{c}_{\text{hi}}, \mathbf{c}_{\text{lo}}$ and an Ajtai lattice authentication tag $\boldsymbol{\tau} \in \mathbb{F}_q^4$.

3 Dual Vandermonde-CRT Verifiable Secret Sharing (VSS)

To distribute a 256-bit Master Key across N custodians such that any T custodians ($2 \le T \le N$) can reconstruct the key while any $T - 1$ shares reveal zero information, CLAT-VSS embeds the 16 key words $\mathbf{m} \in \{0, \ldots, 65535\}^{16} \subset \mathbb{F}_q^{16}$ as the first row $\mathbf{F}_{0,:} = \mathbf{m}^T$ of a degree-$(T-1)$ polynomial coefficient matrix $\mathbf{F} \in \mathbb{F}_q^{T\times 16}$, paired with a uniform random blinding matrix $\mathbf{B} \in \mathbb{F}_q^{T\times 16}$.

3.1 Vandermonde Projection and Chinese Remainder Theorem (CRT) Channel

Each custodian $i \in \{1, \ldots, N\}$ is assigned evaluation point $\alpha_i = i \in \mathbb{F}_q^*$ with Vandermonde vector $\mathbf{v}(\alpha_i) = [1, \alpha_i, \alpha_i^2, \ldots, \alpha_i^{T-1}]^T \in \mathbb{F}_q^T$ and receives the linear projection:

$$\mathbf{y}_i = \mathbf{v}(\alpha_i)^T\mathbf{F} \pmod q, \qquad \mathbf{z}_i = \mathbf{v}(\alpha_i)^T\mathbf{B} \pmod q. \tag{13}$$

In addition, each share coordinate $y_{i,w} \in \mathbb{F}_q$ is decomposed into a **Chinese Remainder Theorem (CRT)** residue pair $(r_{1,w}, r_{2,w})$ modulo the pairwise coprime moduli $m_1 = 65,537$ (the Fermat prime F_4) and $m_2 = 65,521$:

$$r_{1,w} = y_{i,w} \bmod m_1, \qquad r_{2,w} = y_{i,w} \bmod m_2. \tag{14}$$

Since $\gcd(m_1, m_2) = 1$ and $M = m_1 m_2 = 4,294,049,777 > q = 2,147,483,647$, the constructive CRT reconstruction formula uniquely lifts $(r_{1,w}, r_{2,w})$ back to $\mathbb{Z}_M$:

$$\hat{y}_{i,w} = \Big(r_{1,w} m_2 (m_2^{-1} \bmod m_1) + r_{2,w} m_1 (m_1^{-1} \bmod m_2)\Big) \bmod M, \tag{15}$$

where $m_2^{-1} \bmod m_1 = 4,096$ and $m_1^{-1} \bmod m_2 = 61,426$. Any memory bit-flip or transmission fault that alters $y_{i,w}$ without satisfying both coprime residue congruences violates $\hat{y}_{i,w} = y_{i,w}$.

3.2 Dual Lattice and Cyclic-Subgroup Homomorphic Share Validation

To allow each custodian (and the combiner) to verify share authenticity without revealing the master key, CLAT-VSS publishes two complementary homomorphic commitments over the exact same scalar field $\mathbb{F}_q$:

1. **Linear-Algebraic Ajtai/SIS Lattice Commitment ($\mathbf{C_{lat}} \in \mathbb{F}_q^{T\times 8}$):** Using public full-rank matrices $\mathbf{G}, \mathbf{H} \in \mathbb{F}_q^{4\times 8}$, the dealer publishes $\mathbf{C}_{\text{lat}} = \mathbf{F}\mathbf{G}^T + \mathbf{B}\mathbf{H}^T \pmod q$. Custodian i verifies the matrix identity:
$$\mathbf{y}_i \mathbf{G}^T + \mathbf{z}_i \mathbf{H}^T \overset{?}{\equiv} \mathbf{v}(\alpha_i)^T \mathbf{C}_{\text{lat}} \pmod q. \tag{16}$$

2. **Number-Theoretic Pedersen Discrete-Logarithm Commitment ($\mathbf{C_{dlp}} \in \mathbb{G}_q^{T\times 16}$):** We construct the prime modulus $P = 46q + 1 = 98,784,247,763$. By Lagrange's theorem, the multiplicative group $\mathbb{F}_P^*$ of order $P-1 = 46q$ contains a unique cyclic subgroup $\mathbb{G}_q = \{x^{46} \bmod P : x \in \mathbb{F}_P^*\}$ of **exact prime order** $q = 2^{31} - 1$, with verifiably independent generators $g = 2^{46} \bmod P = 34,359,770,408$ and $h = 3^{46} \bmod P = 13,490,067,929$ (satisfying $g^q \equiv h^q \equiv 1 \pmod P$). The dealer publishes:
$$(\mathbf{C}_{\text{dlp}})_{t,w} = g^{\mathbf{F}_{t,w}} \cdot h^{\mathbf{B}_{t,w}} \bmod P \quad \forall t \in \{0,\dots,T-1\},\ w \in \{0,\dots,15\}. \tag{17}$$
Custodian i validates each coordinate w in constant time via:
$$g^{y_{i,w}} \cdot h^{z_{i,w}} \overset{?}{\equiv} \prod_{t=0}^{T-1} \Big((\mathbf{C}_{\text{dlp}})_{t,w}\Big)^{\alpha_i^t \bmod q} \pmod P. \tag{18}$$

Theorem 3.1 (Completeness, Unconditional Hiding, and Dual Binding of CLAT-VSS)**.** *Every honestly generated share $(\mathbf{y}_i, \mathbf{z}_i)$ satisfies both verification equations with probability 1. Furthermore, the public commitments $(\mathbf{C}_{lat}, \mathbf{C}_{dlp})$ leak zero Shannon information about the master key $\mathbf{F}_{0,:}$, and forging a false share $(\tilde{\mathbf{y}}_i, \tilde{\mathbf{z}}_i) \neq (\mathbf{y}_i, \mathbf{z}_i)$ requires simultaneously solving both the Discrete Logarithm Problem $\log_g(h)$ in $\mathbb{G}_q \subset \mathbb{F}_P^*$ and finding a kernel collision in $(\mathbf{G}, \mathbf{H})$ over $\mathbb{F}_q$.*

Proof. Because $\text{ord}_P(g) = \text{ord}_P(h) = q$, exponent arithmetic modulo P is naturally isomorphic to $\mathbb{F}_q$:
$$\prod_{t=0}^{T-1} \Big(g^{F_{t,w}} h^{B_{t,w}}\Big)^{\alpha_i^t \bmod q} \equiv g^{\sum_{t=0}^{T-1} F_{t,w}\alpha_i^t \bmod q} \cdot h^{\sum_{t=0}^{T-1} B_{t,w}\alpha_i^t \bmod q} \equiv g^{y_{i,w}} h^{z_{i,w}} \pmod P. \tag{19}$$
Since $B_{t,w} \sim \mathcal{U}(\mathbb{F}_q)$ is chosen uniformly and independently, $h^{B_{t,w}} \bmod P$ is uniform over $\mathbb{G}_q$, making $\mathbf{C}_{\text{dlp}}$ information-theoretically hiding. Any forged share $\tilde{y}_{i,w} \neq y_{i,w}$ passing the Pedersen check implies $g^{\tilde{y}_{i,w} - y_{i,w}} \equiv h^{z_{i,w} - \tilde{z}_{i,w}} \pmod P$, yielding $\log_g(h) \equiv (\tilde{y}_{i,w} - y_{i,w})(z_{i,w} - \tilde{z}_{i,w})^{-1} \pmod q$. □

4 Side-Channel Mitigations and Vulnerability Window Analysis

To prevent timing, cache, power (SPA/DPA/CPA), and cold-boot/RAM-remnant attacks, CLAT-VSS enforces four hardware-aware defensive layers in C:

Table 1: Side-Channel Attack Vectors and Mathematical/Low-Level Mitigations in CLAT-VSS

Attack Vector	**Vulnerability Window**	**CLAT-VSS Mitigation Mechanism**
Integer Division Timing	Variable CPU cycles in `div/idiv`	Mersenne prime $q = 2^{31} - 1$ bitwise shift-and-add folding (`fq_reduce`); zero division instructions.
Cache-Timing (S-Box)	L1/L2 cache line evictions	Algebraic Euler bijection $S(x) = x^5 \bmod q$ computed in 3 register multiplications; zero lookup tables in RAM.
SPA / Branch Timing	Conditional jumps on secret bits	Constant-time Montgomery powering ladder (`fq_pow_ct`, `fp_pow_ct`) and bitwise multiplexers (`ct_select`).
DPA / CPA Power Traces	Hamming weight of intermediate keys	First-order arithmetic masking splits secret keys and shares into ephemeral random shares $k = k^{(0)} + k^{(1)} \bmod q$ across linear Cauchy/Vandermonde layers.
RAM Remnant / Dump	Plaintext key lifetime in stack/heap	Compiler-barrier volatile scrubbing (`secure_zeroize`) restricts unmasked key lifetime to 272.86 μs.

4.1 First-Order Arithmetic Masking During Decryption and Reconstruction

During both Cauchy MDS inverse transformation and threshold Vandermonde reconstruction, unmasked secrets are never accumulated in CPU registers. Given any T validated shares $\{\mathbf{y}_{i_1}, \ldots, \mathbf{y}_{i_T}\}$, the combiner computes the Lagrange basis weights $\lambda_j = \prod_{k \neq j} \alpha_{i_k} (\alpha_{i_k} - \alpha_{i_j})^{-1} \bmod q$. Before multiplying by λ_j, each share word $y_{i_j,w}$ is split using fresh CSPRNG entropy $m_{j,w} \sim \mathcal{U}(\mathbb{F}_q)$ into two independent arithmetic shares:

$$y^{(0)}_{i_j,w} = m_{j,w}, \qquad y^{(1)}_{i_j,w} = (y_{i_j,w} - m_{j,w}) \bmod q. \tag{20}$$

Two disjoint accumulators $\mathrm{acc}^{(0)}_w = \sum_{j=1}^{T} \lambda_j y^{(0)}_{i_j,w} \bmod q$ and $\mathrm{acc}^{(1)}_w = \sum_{j=1}^{T} \lambda_j y^{(1)}_{i_j,w} \bmod q$ are evaluated separately and recombined ($\mathbf{F}_{0,w} = \mathrm{acc}^{(0)}_w + \mathrm{acc}^{(1)}_w \bmod q$) only at the final output boundary, decorrelating intermediate register Hamming weights from the master key.

5 Empirical Cryptanalysis and Benchmark Verification

The C implementation (`clat_vss.c`, compiled with `gcc -O3 -Wall -Wextra -Werror`) includes an applied **Threshold HSM Master-Key Vault** ($T = 3, N = 5$) and an automated 6-stage cryptanalysis harness evaluating the **Strict Avalanche Criterion (SAC)**, **Golomb's Three Randomness Postulates** (62,400-bit ciphertext stream), **Shannon Byte Entropy**, and a **DudeCT Welch's two-sample t-test** across 4,000 executions comparing Class-0 (all-zero key/plaintext) against Class-1 (maximum Hamming-weight key/plaintext).

References

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Table 2: Empirical Cryptanalysis and Constant-Time Verification Results (`gcc -O3` on x86_64)

Cryptanalytic Metric	**Theoretical Ideal**	**CLAT-VSS Measured**	**Security Verdict**
Strict Avalanche Criterion (SAC)	50.00% bit flip	**49.92**% (1,000 trials)	Passes SAC diffusion bound
Golomb Postulate R1 (Bit Balance)	50.00% ones / zeros	**50.11**% ones / **49.89**% zeros	Zero statistical bias
Golomb Postulate R2 (Runs $L = 1, 2, 3$)	50.0%, 25.0%, 12.5%	**50.1**%, **25.2**%, **12.4**%	Exponential run decay verified
Golomb Postulate R3 (Autocorrelation)	$\lvert C(\tau)\rvert \to 0$ ($\tau \neq 0$)	$\max_\tau \lvert C(\tau)\rvert = \mathbf{0.0071}$	Shift-independent output stream
Ciphertext Shannon Entropy $H(X)$	8.0000 bits/byte	**7.9929** bits/byte	Indistinguishable from uniform
Byzantine 1-Bit Share Tamper Test	100% rejection	**100**% rejected (`rc = -2`)	Dual Lattice+DLP+CRT catch
DudeCT Constant-Time Welch's $\lvert t\rvert$	$\lvert t\rvert < 4.50$	$\lvert\mathbf{t}\rvert = \mathbf{0.45}$ (0.59% Δ)	Statistically constant-time
Plaintext Vulnerability Window	Minimal RAM exposure	**272.86** μs (auto-zeroized)	Sub-millisecond RAM window